SECURITY



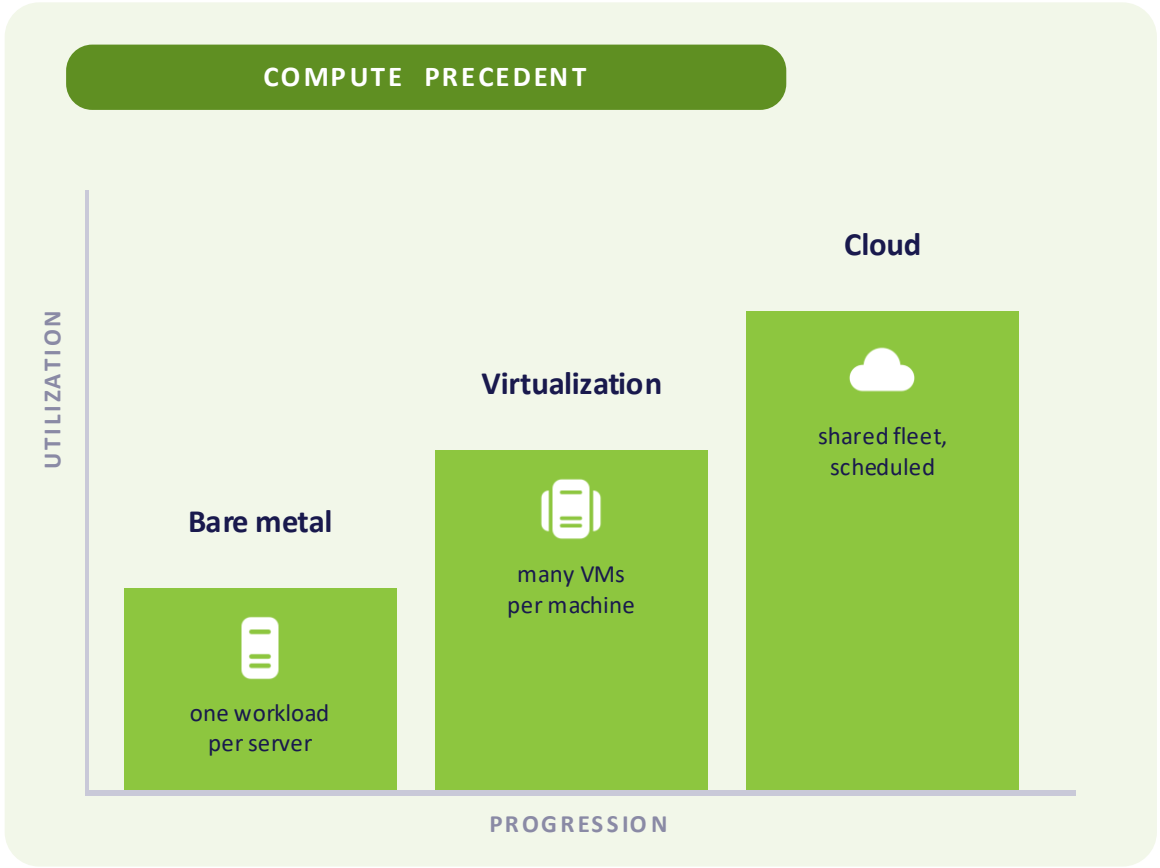
From Securing Links to Securing Pods: Security of Scale-Up Networks

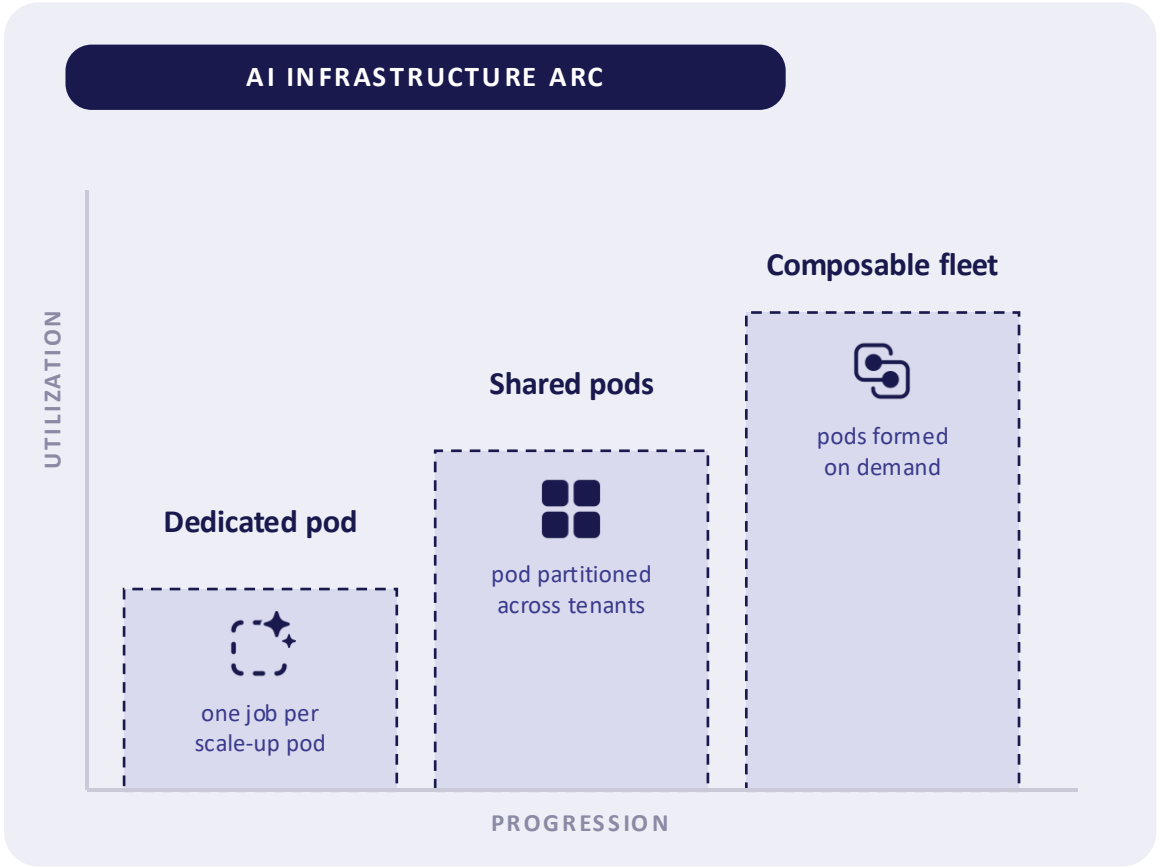
Avdhesh Chhodavdia

Technologist, Astera Labs



Sharing for efficient scaling





Will pod be the new machine and the new unit of sharing?

Agenda

The scale-up pod
One machine, many devices

The claim
Composition matters

Various trust planes
They all contribute

The security model
What must be proven

Operationalize
Conditional lease

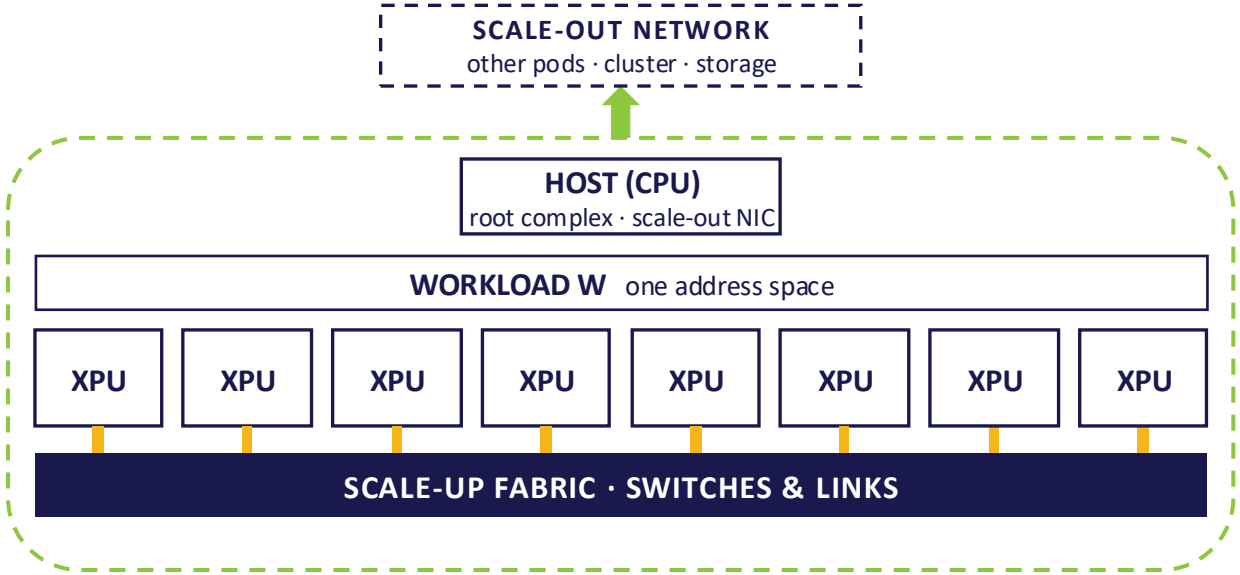
Call to action
Three asks for the group

A scale-up pod executes as one machine

One workload in a single address space.
Traffic is memory-semantic, and the fabric routes, orders and isolates every exchange.

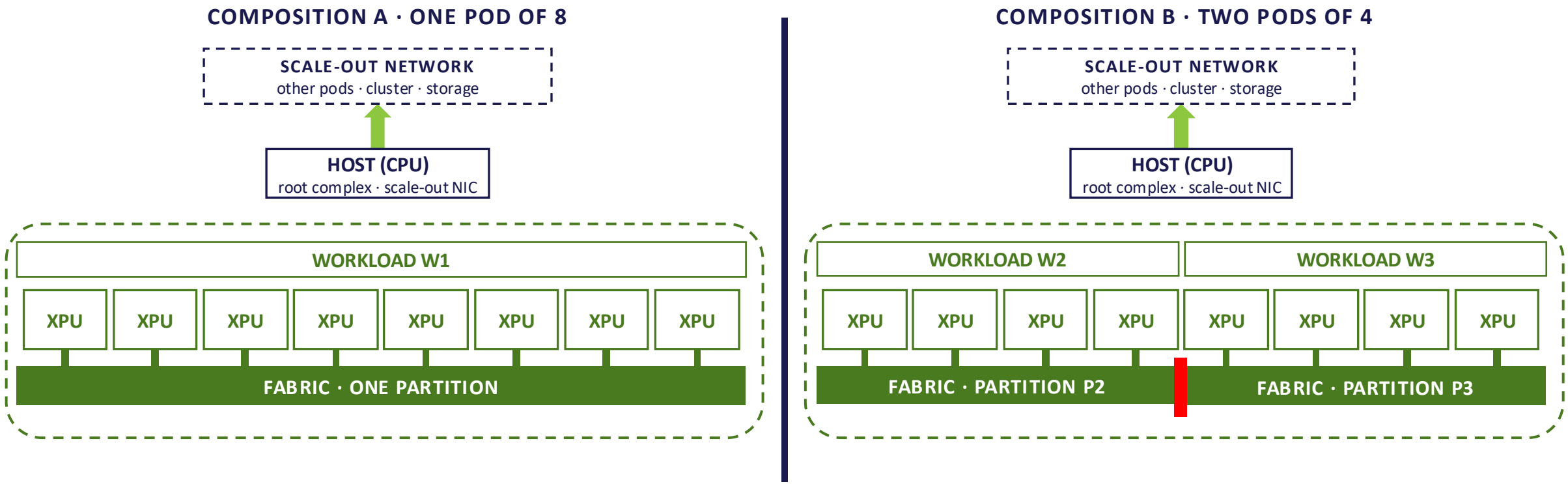
One workload, many accelerators
correctness depends on the pod, not on any single device

The fabric participates in varying degrees
forward, terminate, transform, retain



THE FABRIC IS NOT A BYSTANDER — it participates and hence is part of the security argument

Same devices, different pods



SAME DEVICES, SAME CABLES — membership and mappings decide who is inside which pod

The composition is the unit of assurance

THE CLAIM

Endpoint attestation and protected links are necessary evidence.

In dynamically composed systems these may be insufficient.

The composition is the unit of assurance.

THE SCOPE

Conditional, not universal
fabric state matters where it can invalidate a claimed property

Protocol-agnostic
the same questions across PCIe®, UALink™, CXL®, ESUN

IF FABRIC STATE CAN INVALIDATE THE PROPERTY — the composition is critical to security

Management plane: prove who is in the pod

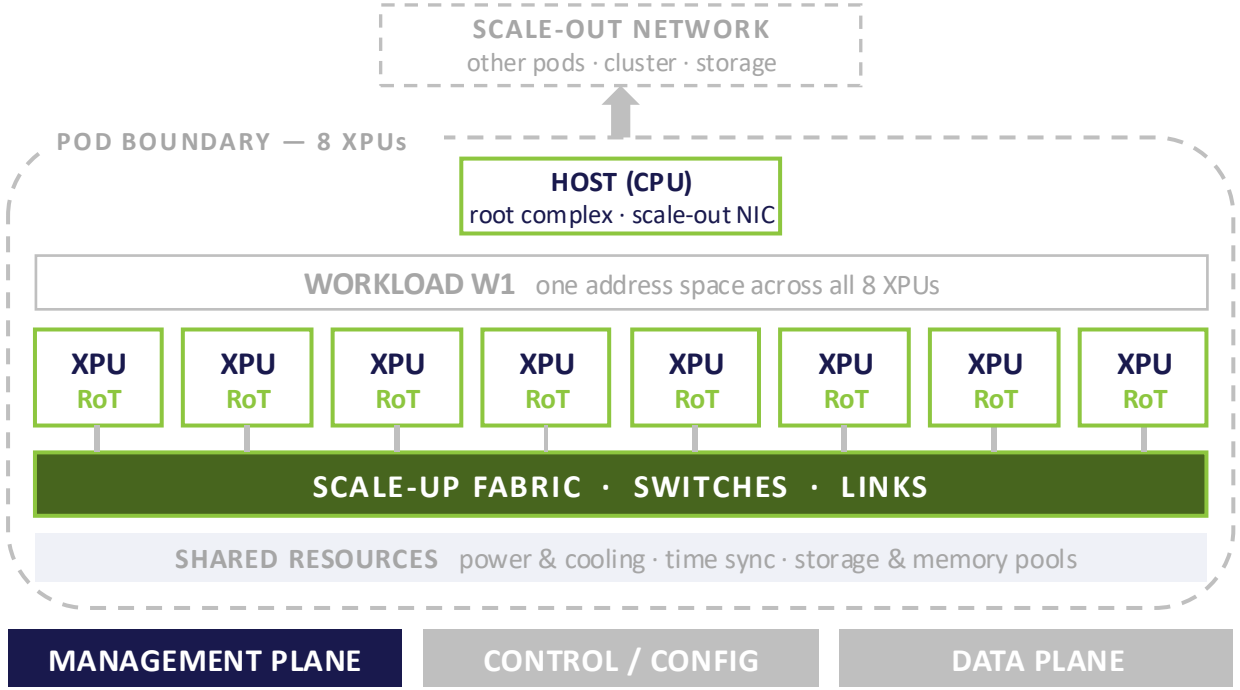
MANAGEMENT PLANE · ASSURANCE OBJECTIVES

Identity
every accelerator, switch, NIC, and controller authenticated

Firmware & static configuration
measured state verified before participation

Inventory
authoritative record of devices, links, versions and lifecycle state

Telemetry & recovery
update, debug and recovery limited to authorized actors



FAILURE a cloned, downgraded or stale component can keep fabric privileges

DESIGN INTENT — no device joins, remains or recovers without fresh evidence and policy authorization

Control plane: the authority that forms the pod

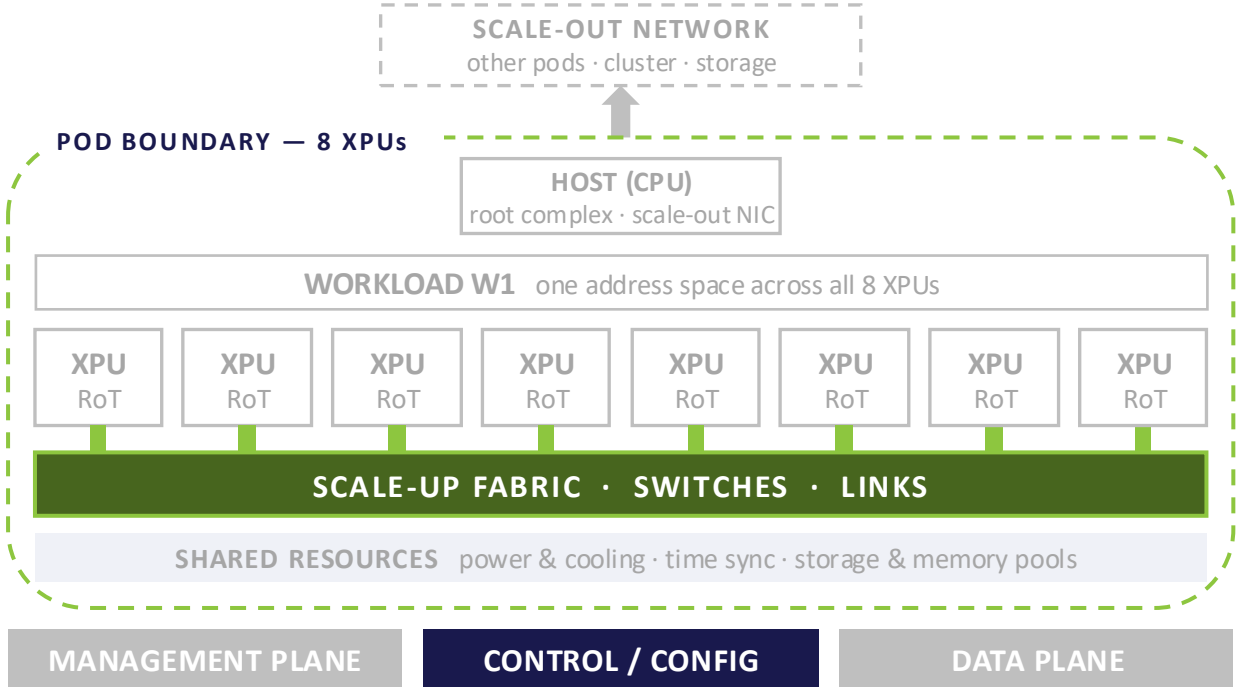
CONTROL / CONFIG PLANE · SECURITY DECISIONS

Discovery & partitioning
which devices and links exist, and who shares a fabric domain

Routing & address maps
who can reach whom

Key policy
which links and contexts are protected

QoS & lifecycle state
cross-tenant impact, and when changes are permitted



FAILURE a reconfiguration can silently change who can reach whom

DESIGN INTENT — every configuration change authorized · replay-protected · auditable · bound to measured participants

Data plane: bind protection to state

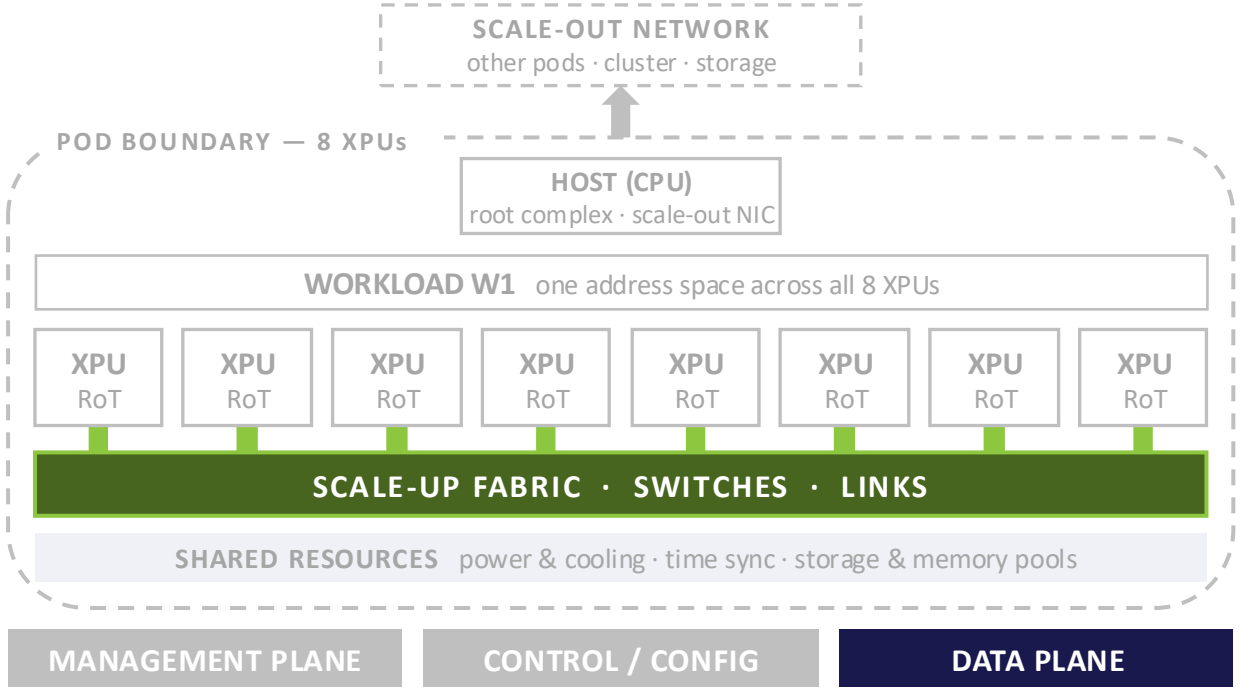
DATA PLANE · RESPONSIBILITIES

Confidentiality & integrity
accelerator-to-accelerator traffic protected

Ordering semantics
ordering and retry/error behavior that collectives and memory operations depend on

Isolation & containment
partitions enforced; faults held inside the boundary

Detection
downgrade, replay, injection, misrouting caught



FAILURE residual tenant state or data can outlive the composition

DESIGN INTENT — encrypted traffic delivered to the right peer, the right partition, as part of approved lifecycle state

A security model answers four questions

1

Claim

What property are we asserting against threats in scope?

2

Dependency

Which plane states can invalidate the claim?

3

Mechanism

What protects it — and where does it terminate?

4

Evidence

Who signs it, and what does it actually mean?

MAKE DEPENDENCY EXPLICIT — define mechanism and collect fresh evidence from each plane

Admission is a lease, not an event

Form
Discover, authenticate, create partitions and paths

Admit
Appraise evidence against intent, then bind the lease

Operate
Watch for anything that changes an assumption

Recover
Retrain, fail over, replace and re-collect evidence

Reprovision
Scrub, reclaim, rebind before reuse

Release
Remove authorization, destroy keys, prove readiness

RUN evidence refreshed, keep claims

RESTRICT reduce the claim, keep running

REVOKE invalidate the lease, destroy keys

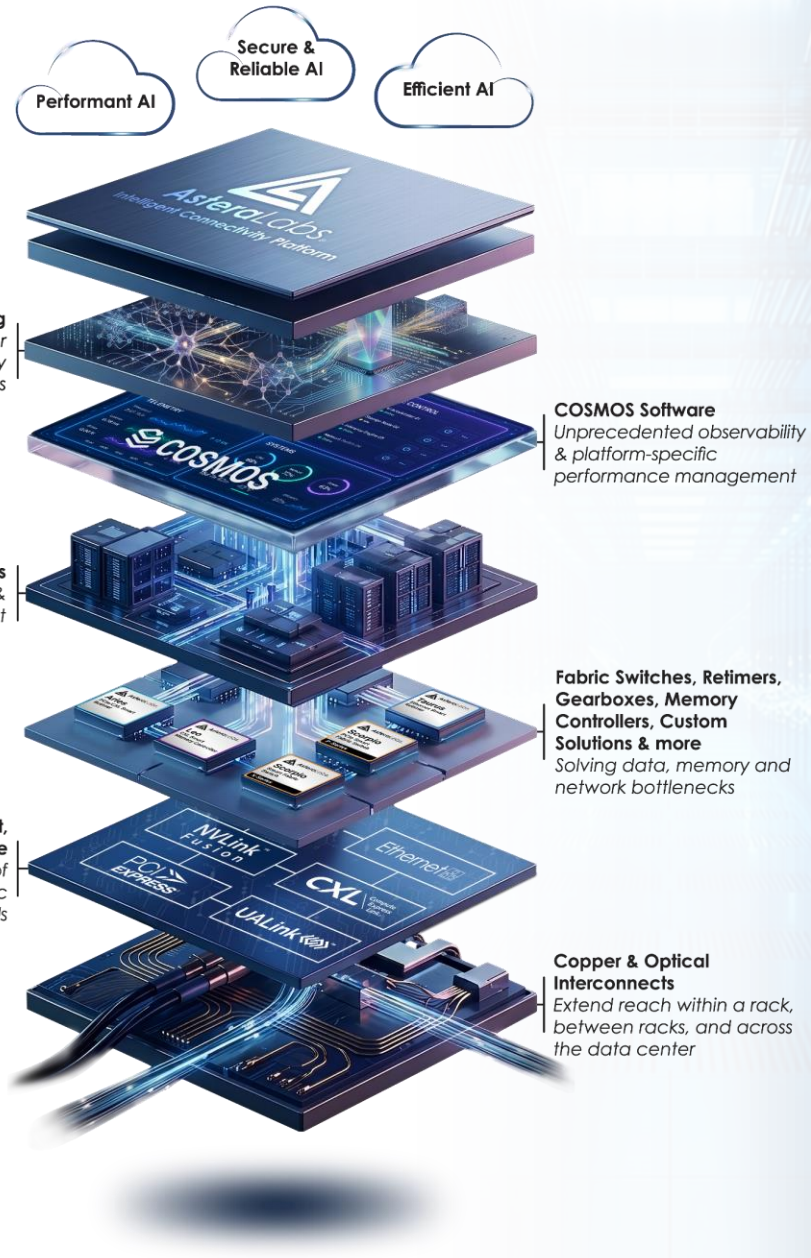
RE-EVALUATE WHEN AN ASSUMPTION CHANGES — otherwise you re-attest constantly, or never

Embracing Open

AI Your Way.

Purpose-built rack-scale connectivity.

any Workload Accelerator



Call to Action

ASK 1	**Define when fabric becomes part of assurance, include evidence-to-component mapping**
ASK 2	**Adopt open protocols for dynamic component binding “unlocked → lock/verify → run”**
ASK 3	**Continue the discussion with Astera Labs at Booth B34**

MAKE POD COMPOSITION A VERIFIABLE SECURITY PROPERTY — not an operational assumption

Thank You!

